

To design an effective and practical BLE-based employee tracking system, it was important to understand existing technologies, systems, and research efforts in the field of attendance monitoring and workforce automation. The following sources and studies helped shape the approach and validated the need for such a solution:

### **1. "Bluetooth Low Energy (BLE) for Industrial Applications" – IEEE Journals**

This study explored how BLE can be applied in real-time industrial tracking scenarios. It confirmed BLE's low power consumption and reliable range (10–30 meters), making it suitable for employee detection at entry/exit points.

### **2. "Time and Attendance Tracking with IoT Devices" – ResearchGate**

This paper discussed the inefficiencies of RFID and biometric systems. It proposed IoT-based models that automate attendance tracking and minimize human intervention, thereby enhancing data reliability and reducing fraud.

### **3. "Firebase as a Scalable Backend for Real-Time Applications" – Firebase Documentation and Developer Blogs**

Firebase is widely used in IoT projects due to its real-time data synchronization, easy API access, and integration with web and mobile platforms. It provides robust support for live dashboards and cloud-hosted data.

### **4. "ESP32 Technical Reference Manual" – Espressif Systems**

The ESP32 is a low-cost, feature-rich microcontroller with integrated BLE and Wi-Fi. Its technical documentation was crucial for designing the firmware logic, enabling it to detect beacon signals and connect to Firebase.

### **5. "IoT-Based Smart Attendance System Using BLE Beacons" – International Journal of Advanced Research in Computer Engineering & Technology (IJARCET)**

This paper presented a system similar to our concept. It emphasized the advantages of using BLE beacons over RFID, including passive detection and higher automation. However, their design lacked real-time data sync and web dashboard integration, which our system includes.

These references collectively justify the technical feasibility and growing relevance of BLE-based attendance systems. They reinforce the direction taken in this project, which combines BLE, ESP32, and Firebase to create a robust, scalable, and real-time employee tracking system.